Bayesian Parameter Estimation

Combining previous information with observed data

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Semester A, 2026/27

Course assessment update

The assessment scheme has been updated:

Assessment	Weight
Assignment	**30%**
Midterm Exam	**10%**
Group Project	30%
Final Exam	30%

Learning objectives

From Week 2 point estimates to uncertainty over parameter values

1. apply Bayes' rule to unknown model parameters;
2. update coin and Gaussian models after observing data;
3. compare maximum likelihood, the most probable posterior value, and posterior prediction;
4. explain how strongly prior information affects an estimate;
5. show how a Gaussian prior creates a penalty on large parameters;
6. identify when a prior helps and when it can mislead.

From likelihood to posterior

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From point estimates to uncertainty

Maximum likelihood gives a single parameter estimate:

$$\hat{\theta}_{\mathrm{ML}} = \arg\max_{\theta} p(\mathcal{D} \mid \theta).$$

But the same point estimate can represent very different levels of uncertainty.

Few observations

Three heads in four flips:

$$\hat{\pi}_{\mathrm{ML}} = \frac{3}{4} = 0.75.$$

More observations

Seventy-five heads in one hundred flips:

$$\hat{\pi}_{\mathrm{ML}} = \frac{75}{100} = 0.75.$$

Both datasets give the same MLE, but should we be equally certain that π is close to 0.75?

Bayesian inference combines two sources of information

Bayesian estimation combines previous knowledge with current observations.

Prior knowledge

Information before observing current data:

$$p(\theta).$$

Examples:

- previous experiments;
- domain knowledge;
- structural constraints.

Observed evidence

Information from current data:

$$p(\mathcal{D} \mid \theta).$$

This is the likelihood used in MLE.

Example: updating belief after new evidence

Suppose we estimate whether a patient has a disease.

1. Prior: belief before observing the test

Previous population studies suggest

$$P(D) = 1\%.$$

This is our initial belief before seeing this patient's test result.

2. Likelihood: evidence provided by the test

The test has known properties:

$$P(+ \mid D) = 99\%, \qquad P(+ \mid \neg D) = 5\%.$$

This describes how likely the observed evidence is under different conditions.

Example: the posterior after updating

After observing a positive test result, we update our belief:

$$P(D \mid +) = \frac{P(+ \mid D)P(D)}{P(+)}.$$

The result is the **posterior**:

$$\text{Posterior} = \text{updated belief after observing data.}$$

In this example,

$$P(D \mid +) = \frac{0.99 \times 0.01}{0.99 \times 0.01 + 0.05 \times 0.99} \approx 16.7\%.$$

The posterior is higher than the prior because the test is informative, but it is not equal to the test sensitivity.

Bayes' rule: updating beliefs after observing data

Bayes' rule describes how beliefs change after observing new data:

$$p(\theta \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \theta)p(\theta)}{p(\mathcal{D})}.$$

Each term has a different role:

- $p(\theta)$: prior belief before observing current data;
- $p(\mathcal{D} \mid \theta)$: likelihood of the observed data under each parameter value;
- $p(\theta \mid \mathcal{D})$: posterior belief after updating with data.

Since $p(\mathcal{D})$ only normalizes the distribution,

$$p(\theta \mid \mathcal{D}) \propto p(\mathcal{D} \mid \theta)p(\theta).$$

Bayesian inference updates a distribution over parameters, rather than directly selecting one parameter value.

A posterior distribution contains more than one estimate

After Bayesian updating, we obtain a distribution:

$$p(\theta \mid \mathcal{D}).$$

This distribution describes:

- which parameter values are plausible;
- how uncertain the estimate remains;
- how much information the data provide.

A point estimate keeps only one summary of this distribution.

Main idea

Two models may have the same posterior mean but very different uncertainty. The full posterior preserves information about confidence.

A posterior distribution can be used in different ways

After Bayesian updating, we have

$$p(\theta \mid \mathcal{D}).$$

This distribution can support different tasks.

Parameter estimation

We may summarize the posterior by a single value, such as its mean:

$$\hat{\theta}_{\text{mean}} = \mathbb{E}[\theta \mid \mathcal{D}].$$

Future prediction

We may instead predict a new observation by averaging over parameter uncertainty:

$$p(x_* \mid \mathcal{D}) = \int p(x_* \mid \theta) p(\theta \mid \mathcal{D}) d\theta.$$

Why prediction should average over uncertainty

A parameter estimate treats the unknown parameter as a single value:

$$x_* \sim p(x_* \mid \hat{\theta}).$$

However, after observing data, we are still uncertain about θ:

$$p(\theta \mid \mathcal{D}).$$

Different parameter values imply different predictions:

$$p(x_* \mid \theta_1) \neq p(x_* \mid \theta_2).$$

Bayesian prediction combines all possible parameter values:

$$p(x_* \mid \mathcal{D}) = \int p(x_* \mid \theta) p(\theta \mid \mathcal{D}) d\theta.$$

Beta–Bernoulli estimation

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Why can we use a Beta distribution for a coin probability?

In a Bernoulli model, the true success probability π is fixed but unknown.
Bayesian inference represents our uncertainty about this unknown parameter:

$$\pi \sim p(\pi).$$

Since

$$0 \leq \pi \leq 1,$$

the prior distribution should also be defined on the interval $[0, 1]$.
The Beta distribution is a flexible choice:

$$p(\pi) \propto \pi^{\alpha-1}(1-\pi)^{\beta-1}.$$

Different values of α and β represent different prior beliefs about possible coin probabilities.

The Beta distribution does not describe a changing coin; it describes our uncertainty about an unknown coin probability.

Why is Beta a conjugate prior for Bernoulli data?

Bayesian updating combines prior information and observed data:

$$p(\pi \mid \mathcal{D}) \propto p(\mathcal{D} \mid \pi) p(\pi).$$

For Bernoulli observations with m successes and $n - m$ failures:

$$p(\mathcal{D} \mid \pi) \propto \pi^{m} (1 - \pi)^{n-m}.$$

The Beta prior has the form:

$$p(\pi) \propto \pi^{\alpha-1} (1 - \pi)^{\beta-1}.$$

Multiplying them gives:

$$p(\pi \mid \mathcal{D}) \propto \pi^{\alpha+m-1} (1 - \pi)^{\beta+n-m-1}.$$

This is still a Beta distribution:

$$\boxed{p(\pi \mid \mathcal{D}) = \mathrm{Beta}(\alpha + m, \beta + n - m)}.$$

A conjugate prior keeps the posterior in the same family, making Bayesian updating easy to compute.

Understanding the Beta prior: mean and concentration

A Beta prior has two parameters α and β. Its mean and variance are

$$\mu_0 = \mathbb{E}[\pi] = \frac{\alpha}{\alpha+\beta}, \qquad \mathrm{Var}(\pi) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}.$$

To make these parameters easier to interpret, define the concentration

$$\kappa_0 = \alpha + \beta.$$

Then

$$\alpha = \mu_0\kappa_0, \qquad \beta = (1-\mu_0)\kappa_0.$$

Substituting these into the variance gives

$$\mathrm{Var}(\pi) = \frac{\mu_0(1-\mu_0)}{\kappa_0+1}.$$

The mean μ_0 determines where the prior is centered, while the concentration κ_0 determines how tightly it is concentrated.

Different Beta priors express different beliefs

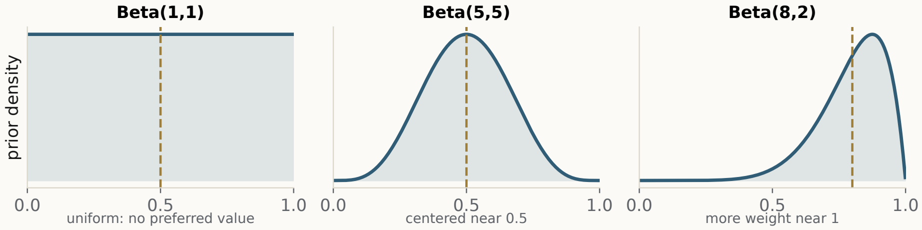
Priors with different parameters can represent very different information:

- $\mathrm{Beta}(1,1)$: uniform over $[0,1]$;
- $\mathrm{Beta}(10,10)$: centered at 0.5 and more concentrated;
- $\mathrm{Beta}(20,5)$: favors larger success probabilities.

Two priors can even have the same mean but different certainty:

$$\mathrm{Beta}(2,2) \qquad \text{vs.} \qquad \mathrm{Beta}(20,20).$$

Both have mean 0.5, but the second places much more probability near 0.5.



Beta–Bernoulli updating: what comes from the data?

Suppose

$$x_1, \ldots, x_n \sim \text{Bernoulli}(\pi), \qquad m = \sum_{i=1}^{n} x_i.$$

Here, m is the number of successes and $n - m$ is the number of failures. The likelihood is

$$p(\mathcal{D} \mid \pi) = \pi^m (1 - \pi)^{n-m}.$$

The Beta prior has the form

$$p(\pi) \propto \pi^{\alpha - 1} (1 - \pi)^{\beta - 1}.$$

Bayesian updating multiplies these two sources of information:

$$p(\pi \mid \mathcal{D}) \propto p(\mathcal{D} \mid \pi) p(\pi).$$

Beta–Bernoulli updating: deriving the posterior

Substituting likelihood and prior gives

$$p(\pi \mid \mathcal{D}) \propto \pi^m (1-\pi)^{n-m} \pi^{\alpha-1} (1-\pi)^{\beta-1}.$$

Collecting equal powers,

$$p(\pi \mid \mathcal{D}) \propto \pi^{\alpha+m-1} (1-\pi)^{\beta+n-m-1}.$$

This is again the form of a Beta distribution:

$$\boxed{\pi \mid \mathcal{D} \sim \mathrm{Beta}(\alpha+m, \beta+n-m)}.$$

The update therefore has a simple interpretation:

$$\alpha \leftarrow \alpha + m, \qquad \beta \leftarrow \beta + (n-m).$$

Example: how prior information changes the estimate

Suppose the prior belief is

$$\pi \sim \mathrm{Beta}(10, 10).$$

The prior mean is

$$\mathbb{E}[\pi] = \frac{10}{10 + 10} = 0.5.$$

We then observe 8 heads in 10 flips:

$$m = 8, \qquad n - m = 2.$$

The posterior parameters are updated by adding observations:

$$\pi \,|\, \mathcal{D} \sim \mathrm{Beta}(10 + 8, 10 + 2) = \mathrm{Beta}(18, 12).$$

Example: comparing MLE and Bayesian estimation

For the same 8 heads in 10 flips, maximum likelihood gives

$$\hat{\pi}_{\mathrm{ML}} = \frac{8}{10} = 0.8.$$

With the $\mathrm{Beta}(10, 10)$ prior, the posterior is

$$\pi \mid \mathcal{D} \sim \mathrm{Beta}(18, 12),$$

so the posterior mean is

$$\mathbb{E}[\pi \mid \mathcal{D}] = \frac{18}{30} = 0.6.$$

MLE uses only the current observations, whereas the posterior mean also reflects the prior belief centered at 0.5.

The data pull the Bayesian estimate toward 0.8, while the prior prevents ten observations from dominating completely.

Posterior mean: exactly how are prior and data combined?

For the Beta posterior,

$$\mathbb{E}[\pi \mid \mathcal{D}] = \frac{\alpha + m}{\alpha + \beta + n}.$$

Rearranging,

$$\mathbb{E}[\pi \mid \mathcal{D}] = \underbrace{\frac{\alpha + \beta}{\alpha + \beta + n}}_{\text{prior weight}} \underbrace{\frac{\alpha}{\alpha + \beta}}_{\text{prior mean}} + \underbrace{\frac{n}{\alpha + \beta + n}}_{\text{data weight}} \underbrace{\frac{m}{n}}_{\text{data mean}} .$$

Therefore,

$$\text{posterior mean} = w_{\text{prior}} \times \text{prior mean} + w_{\text{data}} \times \text{data mean},$$

where

$$w_{\text{prior}} + w_{\text{data}} = 1.$$

Prior strength matters even when the prior mean is the same

Consider the same data:

$$m = 8, \qquad n = 10,$$

but two priors with the same mean 0.5.
A weaker prior gives

$$\mathrm{Beta}(2, 2) \to \mathrm{Beta}(10, 4), \qquad \mathbb{E}[\pi \mid \mathcal{D}] = \frac{10}{14} \approx 0.714.$$

A stronger prior gives

$$\mathrm{Beta}(20, 20) \to \mathrm{Beta}(28, 22), \qquad \mathbb{E}[\pi \mid \mathcal{D}] = \frac{28}{50} = 0.56.$$

Both priors start at the same mean, but they react differently to the same observations.

A prior contains both a preferred location and a strength of belief.

From posterior distributions to point estimates

Bayesian inference gives a full posterior distribution:

$$p(\pi \mid \mathcal{D}).$$

Sometimes we still need one representative parameter value. Two common choices are:

- **posterior mean**: the average value under the posterior;
- **maximum a posteriori (MAP)**: the value where the posterior density is highest.

The MAP estimate is therefore

$$\hat{\pi}_{\mathrm{MAP}} = \arg\max_{\pi} p(\pi \mid \mathcal{D}).$$

In other words, MAP selects the mode of the posterior distribution.

MAP reduces the full posterior distribution to its highest-density point.

MAP connects Bayesian estimation with maximum likelihood

Maximum likelihood selects the parameter that best explains the data:

$$\hat{\pi}_{\mathrm{ML}} = \arg\max_{\pi} p(\mathcal{D} \mid \pi).$$

MAP considers both the data and prior knowledge:

$$\hat{\pi}_{\mathrm{MAP}} = \arg\max_{\pi} p(\pi \mid \mathcal{D}).$$

Using Bayes' rule:

$$p(\pi \mid \mathcal{D}) \propto p(\mathcal{D} \mid \pi) p(\pi).$$

Therefore,

$$\hat{\pi}_{\mathrm{MAP}} = \arg\max_{\pi} \left[\log p(\mathcal{D} \mid \pi) + \log p(\pi)\right].$$

MLE uses only evidence from data; MAP adds a preference encoded by the prior.

MAP estimation for Beta–Bernoulli

After Bayesian updating,

$$\pi \mid \mathcal{D} \sim \mathrm{Beta}(\alpha + m, \beta + n - m).$$

MAP selects the mode of this posterior distribution. Its log-density, ignoring constants, is

$$(\alpha + m - 1) \log \pi + (\beta + n - m - 1) \log(1 - \pi).$$

If

$$\alpha + m > 1, \qquad \beta + n - m > 1,$$

the posterior has an interior mode. Differentiating gives

$$\boxed{\hat{\pi}_{\mathrm{MAP}} = \frac{\alpha + m - 1}{\alpha + \beta + n - 2}}.$$

If either condition fails, the mode may occur at the boundary $\pi = 0$ or $\pi = 1$.

MAP and posterior mean answer different questions

The posterior mean is

$$\mathbb{E}[\pi \mid \mathcal{D}] = \frac{\alpha + m}{\alpha + \beta + n}.$$

The MAP estimate is

$$\hat{\pi}_{\mathrm{MAP}} = \frac{\alpha + m - 1}{\alpha + \beta + n - 2}.$$

They are different because they summarize the posterior in different ways:

- posterior mean averages over all possible parameter values;
- MAP chooses the single most probable parameter value.

For symmetric posteriors they may be close, but for skewed posteriors they can differ substantially.

Example: MAP versus posterior mean

Consider a skewed posterior:

$$\pi \mid \mathcal{D} \sim \mathrm{Beta}(2, 5).$$

The posterior mean is

$$\mathbb{E}[\pi \mid \mathcal{D}] = \frac{2}{2+5} = \frac{2}{7} \approx 0.286.$$

The MAP estimate is

$$\hat{\pi}_{\mathrm{MAP}} = \frac{2-1}{2+5-2} = 0.2.$$

The difference occurs because the posterior is asymmetric.

There is no universally best posterior summary; the appropriate choice depends on the task.

What does posterior prediction mean?

To understand the idea, first consider a toy example in which π can take only two possible values:

$$P(\pi = 0.2 \mid \mathcal{D}) = 0.3, \qquad P(\pi = 0.6 \mid \mathcal{D}) = 0.7.$$

If $\pi = 0.2$, the probability of success on the next trial is

$$P(x_* = 1 \mid \pi = 0.2) = 0.2.$$

If $\pi = 0.6$, it is

$$P(x_* = 1 \mid \pi = 0.6) = 0.6.$$

Because we do not know which parameter value is correct, we weight the two predictions by their posterior probabilities:

$$P(x_* = 1 \mid \mathcal{D}) = 0.3(0.2) + 0.7(0.6) = 0.48.$$

Posterior prediction combines predictions from plausible parameter values according to their posterior weights.

Posterior prediction for a continuous parameter

For a Beta posterior, the weighted sum becomes an integral. For a future Bernoulli observation,

$$P(x_* = 1 \mid \pi) = \pi.$$

Averaging over the posterior distribution of π gives

$$P(x_* = 1 \mid \mathcal{D}) = \int_0^1 P(x_* = 1 \mid \pi) p(\pi \mid \mathcal{D}) \, d\pi.$$

Substituting $P(x_* = 1 \mid \pi) = \pi$,

$$P(x_* = 1 \mid \mathcal{D}) = \int_0^1 \pi p(\pi \mid \mathcal{D}) \, d\pi.$$

This integral is exactly the posterior mean:

$$\boxed{P(x_* = 1 \mid \mathcal{D}) = \mathbb{E}[\pi \mid \mathcal{D}]}.$$

Example: Bayesian prediction avoids extreme decisions

Suppose we observe five failures: $m = 0, n = 5$. Maximum likelihood gives

$$\hat{\pi}_{\mathrm{ML}} = \frac{0}{5} = 0.$$

A plug-in prediction would conclude:

$$P(x_* = 1 \mid \hat{\pi}_{\mathrm{ML}}) = 0.$$

With a uniform prior

$$\pi \sim \mathrm{Beta}(1, 1),$$

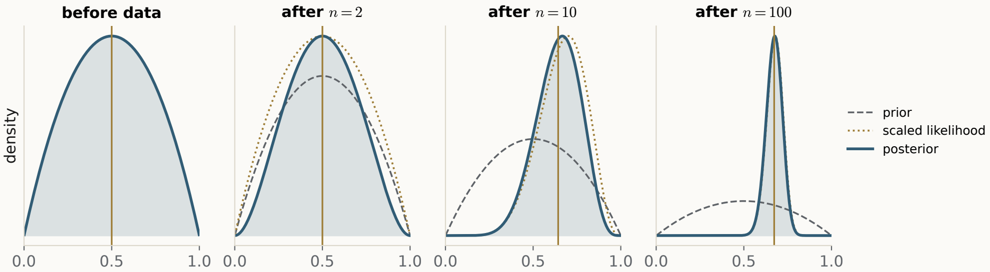
the posterior becomes

$$\pi \mid \mathcal{D} \sim \mathrm{Beta}(1, 6).$$

The Bayesian predictive probability is

$$P(x_* = 1 \mid \mathcal{D}) = \frac{1}{7} \approx 0.143.$$

Visualizing sequential Bayesian updating



Sequential updating repeats the same process:

$$\text{posterior} \rightarrow \text{new prior} \rightarrow \text{new posterior.}$$

- Early observations can move the posterior substantially.
- More data make the posterior concentrate.
- A fixed prior becomes less influential as data accumulate.

Gaussian conjugacy

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Why learn Gaussian conjugacy?

Beta–Bernoulli showed Bayesian updating for binary observations:

$$\text{success/failure data} \to \text{updated belief about } \pi.$$

Many real problems instead involve continuous measurements, such as experimental measurements, sensor readings, or test scores. We may have two different estimates of the same unknown mean:

$$\underbrace{\mu_0}_{\text{previous knowledge}} \qquad \text{and} \qquad \underbrace{\bar{x}}_{\text{new data}} \quad .$$

The key question is

$$\boxed{\text{Which source should we trust more?}}$$

Gaussian conjugacy combines continuous information according to how uncertain each source is.

Problem setup: what are we trying to estimate?

Suppose continuous observations satisfy

$$x_i \mid \mu \sim \mathcal{N}(\mu, \sigma^2).$$

Here, μ is unknown, while σ^2 is assumed known for now. The known σ^2 represents the noise level of individual observations and may come from previous experiments, calibration, or historical data.

- μ: unknown population mean we want to estimate;
- σ^2: known observation variance that measures how noisy individual measurements are;
- $x_1, \ldots, x_n$: current observed measurements used to learn about μ.

In this section, we focus only on uncertainty about μ.

We treat σ^2 as known so that we can isolate the main idea: how Bayesian inference combines prior knowledge and current data to estimate μ.

What exactly is a Gaussian prior?

The true mean μ is fixed but unknown. Before observing the current data, suppose previous knowledge suggests that μ is likely to be near μ_0. We represent this uncertainty by

$$\mu \sim \mathcal{N}(\mu_0, \tau_0^2).$$

This is a **Gaussian prior** over the unknown parameter μ. Its parameters have clear meanings:

$$\underbrace{\mu_0}_{\text{prior best guess}} \qquad \underbrace{\tau_0^2}_{\text{uncertainty in that guess}} \qquad .$$

A smaller τ_0^2 means stronger prior confidence; a larger τ_0^2 means weaker prior knowledge.

The prior describes uncertainty about μ, not variability of the observed measurements.

What do the current data tell us about μ?

For n independent Gaussian observations, the likelihood can be written as

$$p(\mathcal{D} \,|\, \mu) \propto \exp\left[-\frac{n}{2\sigma^2}(\mu - \bar{x})^2\right].$$

Viewed as a function of μ, this has a Gaussian shape:

$$p(\mathcal{D} \,|\, \mu) \propto \exp\left[-\frac{(\mu - \bar{x})^2}{2(\sigma^2/n)}\right].$$

Therefore the data provide:

$$\underbrace{\bar{x}}_{\text{preferred value}} \qquad \underbrace{\sigma^2/n}_{\text{uncertainty}} .$$

Since

$$\mathrm{Var}(\bar{x}) = \frac{\sigma^2}{n},$$

more observations make the data estimate of μ more reliable.

Now we have two sources of information

The prior provides

$$\text{center } = \mu_0, \qquad \text{uncertainty } = \tau_0^2.$$

The current data provide

$$\text{center } = \bar{x}, \qquad \text{uncertainty } = \frac{\sigma^2}{n}.$$

These two centers may disagree:

$$\mu_0 \neq \bar{x}.$$

But disagreement alone does not tell us which one to trust. We must also ask:

$$\boxed{\text{How uncertain is each source?}}$$

Precision turns uncertainty into weight

Variance measures uncertainty. Its inverse is called **statistical precision**:

$$\text{precision} = \frac{1}{\text{variance}}.$$

The prior precision is

$$\lambda_0 = \frac{1}{\tau_0^2}.$$

The data uncertainty is σ^2/n, so the data precision is

$$\lambda_D = \frac{1}{\sigma^2/n} = \frac{n}{\sigma^2}.$$

Therefore:

- small variance $\Rightarrow$ high precision $\Rightarrow$ more reliable;
- large variance $\Rightarrow$ low precision $\Rightarrow$ less reliable.

Gaussian conjugacy: combining prior and data

Bayes' rule gives

$$p(\mu \,|\, \mathcal{D}) \propto p(\mathcal{D} \,|\, \mu)p(\mu).$$

Using the two Gaussian-shaped information sources,

$$p(\mu \,|\, \mathcal{D}) \propto \exp\left[-\frac{1}{2}\left(\lambda_D(\mu - \bar{x})^2 + \lambda_0(\mu - \mu_0)^2\right)\right].$$

The exponent is still quadratic in μ, so the posterior is also Gaussian:

$$\boxed{\mu \,|\, \mathcal{D} \sim \mathcal{N}(\mu_n, \tau_n^2)}.$$

This is the conjugacy:

$$\text{Gaussian prior} + \text{Gaussian likelihood} \to \text{Gaussian posterior.}$$

How do we identify the posterior Gaussian?

We already know $p(\mu \,|\, \mathcal{D})$ is Gaussian, but we still need to find its mean and variance. A Gaussian in μ has the standard form

$$p(\mu \,|\, \mathcal{D}) \propto \exp\left[-\frac{1}{2}\lambda_n(\mu - \mu_n)^2\right],$$

where

$$\mu_n = \text{posterior mean}, \qquad \lambda_n = \frac{1}{\tau_n^2} = \text{posterior precision}.$$

Therefore, our goal is to rewrite the two quadratic terms as one squared term:

$$\lambda_D(\mu - \bar{x})^2 + \lambda_0(\mu - \mu_0)^2 = \lambda_n(\mu - \mu_n)^2 + C.$$

Completing the square gives the posterior parameters

Expand only the terms involving μ:

$$\lambda_D(\mu - \bar{x})^2 + \lambda_0(\mu - \mu_0)^2 = (\lambda_D + \lambda_0)\mu^2 - 2(\lambda_D \bar{x} + \lambda_0 \mu_0)\mu + C.$$

Compare this with

$$\lambda_n(\mu - \mu_n)^2 = \lambda_n \mu^2 - 2\lambda_n \mu_n \mu + C.$$

Matching the coefficients gives

$$\boxed{\lambda_n = \lambda_0 + \lambda_D}$$

and

$$\boxed{\mu_n = \frac{\lambda_0 \mu_0 + \lambda_D \bar{x}}{\lambda_0 + \lambda_D}}.$$

Since $\lambda_n = 1/\tau_n^2$,

$$\boxed{\tau_n^2 = \frac{1}{\lambda_0 + \lambda_D}}.$$

Why is the posterior mean a weighted average?

Rewrite the posterior mean as

$$\mu_n = \underbrace{\frac{\lambda_0}{\lambda_0 + \lambda_D}}_{w_0} \mu_0 + \underbrace{\frac{\lambda_D}{\lambda_0 + \lambda_D}}_{w_D} \bar{x},$$

where

$$w_0 + w_D = 1.$$

Therefore:

- small $\tau_0^2 \Rightarrow$ strong prior $\Rightarrow$ larger prior weight;
- large $n \Rightarrow$ more reliable data $\Rightarrow$ larger data weight;
- large $\sigma^2 \Rightarrow$ noisier observations $\Rightarrow$ smaller data weight.

The more precise source pulls the posterior mean closer to itself.

Example: which source receives more weight?

Suppose the prior is

$$\mu \sim \mathcal{N}(10, 4).$$

Hence

$$\mu_0 = 10, \qquad \lambda_0 = \frac{1}{4} = 0.25.$$

Now suppose

$$n = 4, \qquad \sigma^2 = 4, \qquad \bar{x} = 12.$$

The data precision is

$$\lambda_D = \frac{n}{\sigma^2} = 1.$$

Therefore the weights are

$$w_0 = \frac{0.25}{1.25} = 0.2, \qquad w_D = \frac{1}{1.25} = 0.8.$$

The data receive more weight because they are more precise.

Example: computing the posterior

Using the previous weights,

$$\mu_n = 0.2(10) + 0.8(12) = 11.6.$$

The posterior precision is

$$\lambda_n = 0.25 + 1 = 1.25,$$

so the posterior variance is

$$\tau_n^2 = \frac{1}{1.25} = 0.8.$$

Therefore,

$$\boxed{\mu \mid \mathcal{D} \sim \mathcal{N}(11.6, 0.8)}.$$

The posterior moves from the prior mean 10 toward the data mean 12, while becoming more certain.

Connection to maximum likelihood

For this Gaussian model, maximum likelihood gives

$$\hat{\mu}_{\mathrm{ML}} = \bar{x}.$$

The Bayesian posterior mean is

$$\mu_n = \frac{\lambda_0 \mu_0 + \lambda_D \bar{x}}{\lambda_0 + \lambda_D}.$$

If the prior becomes very weak,

$$\tau_0^2 \to \infty \qquad \Rightarrow \qquad \lambda_0 \to 0.$$

Then

$$\mu_n \to \bar{x} = \hat{\mu}_{\mathrm{ML}}.$$

The same happens when the amount of data becomes very large.

From estimating μ to predicting a new observation

So far, we have used the observed data to learn about the unknown mean:

$$\mu \,|\, \mathcal{D} \sim \mathcal{N}(\mu_n, \tau_n^2).$$

But in many applications, estimating μ is not the final goal. We want to predict what we may observe next. Let

$$x_* = \text{a new observation from the same population.}$$

For example, after estimating the average measurement from existing patients, x_* could be the measurement from a new patient. If the true mean μ were known, the new observation would follow

$$x_* \,|\, \mu \sim \mathcal{N}(\mu, \sigma^2).$$

Our prediction problem is therefore

$$\boxed{\text{What is } p(x_* \,|\, \mathcal{D}) \text{ when } \mu \text{ is still uncertain?}}$$

Why is predicting x_* different from estimating μ?

The posterior

$$\mu \mid \mathcal{D} \sim \mathcal{N}(\mu_n, \tau_n^2)$$

describes uncertainty about the population mean. A new observation can be written as

$$x_* = \mu + \epsilon_*, \qquad \epsilon_* \sim \mathcal{N}(0, \sigma^2).$$

Therefore x_* is uncertain for two different reasons:

- we do not know the true mean μ exactly;
- even if μ were known, individual observations would still vary around it.

These correspond to

$$\underbrace{\tau_n^2}_{\text{uncertainty about } \mu} \qquad \text{and} \qquad \underbrace{\sigma^2}_{\text{variation of a new observation}} .$$

Estimating the mean and predicting a new observation are different tasks.

How do we obtain the posterior predictive distribution?

Because the true μ is unknown, we should not make a prediction using only one assumed value of μ. Instead, we consider every possible μ and weight it by its posterior probability:

$$p(x_* \mid \mathcal{D}) = \int p(x_* \mid \mu) p(\mu \mid \mathcal{D}) \, d\mu.$$

Here,

$$p(x_* \mid \mu)$$

tells us how a new observation behaves if μ were known, while

$$p(\mu \mid \mathcal{D})$$

tells us how plausible each possible value of μ is after observing the data. Since both distributions are Gaussian, the resulting predictive distribution is

$$\boxed{p(x_* \mid \mathcal{D}) = \mathcal{N}(x_* \mid \mu_n, \sigma^2 + \tau_n^2)}.$$

Example: estimating the mean versus predicting a new value

From our previous example, the posterior was

$$\mu \mid \mathcal{D} \sim \mathcal{N}(11.6, 0.8).$$

This means our best estimate of the population mean is 11.6, with remaining parameter uncertainty 0.8. The observation variance was $\sigma^2 = 4$. For a new observation x_*, the predictive distribution is therefore

$$x_* \mid \mathcal{D} \sim \mathcal{N}(11.6, 4 + 0.8) = \mathcal{N}(11.6, 4.8).$$

Notice the difference:

$$\underbrace{0.8}_{\text{uncertainty about the mean}} < \underbrace{4.8}_{\text{uncertainty about a new observation}}.$$

A future observation is more uncertain than the estimated mean because it contains both parameter uncertainty and new observation noise.

MAP and penalties on large parameters

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Why do we need regularization in regression?

Maximum likelihood chooses regression weights that fit the observed data:

$$\hat{\boldsymbol{w}}_{\mathrm{ML}} = \arg\min_{\boldsymbol{w}} \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{w}\|_2^2.$$

But a good training fit does not always imply reliable parameters. Problems can occur when:

- the dataset is small relative to the number of features;
- several features contain very similar information;
- noise causes the fitted weights to become unnecessarily large.

In such cases, slightly different datasets may produce very different weight estimates.

Regularization

Add a preference for simpler or more plausible parameter values instead of optimizing data fit alone.

From one Gaussian prior to a prior on many weights

Linear regression has several unknown parameters:

$$\boldsymbol{w} = (w_1, \ldots, w_d)^\top.$$

We place the same simple Gaussian prior on each coefficient:

$$w_j \sim \mathcal{N}(0, \tau^2), \qquad j = 1, \ldots, d.$$

Assuming these prior beliefs are independent, the joint prior becomes

$$\boxed{\boldsymbol{w} \sim \mathcal{N}(\boldsymbol{0}, \tau^2 \boldsymbol{I})}.$$

Here:

- mean 0: each weight is expected to be near zero before seeing the data;
- variance τ^2: controls how strongly large weights are discouraged;
- $\boldsymbol{I}$: all weights use the same prior variance and have no prior correlation.

From individual weight priors to the joint prior

From the previous slide, each regression weight has the prior $w_j \sim \mathcal{N}(0, \tau^2)$ with density:

$$p(w_j) = \frac{1}{\sqrt{2\pi\tau^2}} \exp\left(-\frac{w_j^2}{2\tau^2}\right), j = 1, \ldots, d.$$

Assuming the weights are independent under the prior, their joint density is the product

$$p(\boldsymbol{w}) = \prod_{j=1}^{d} p(w_j).$$

Therefore,

$$p(\boldsymbol{w}) = (2\pi\tau^2)^{-d/2} \exp\left[-\frac{1}{2\tau^2}\sum_{j=1}^{d} w_j^2\right] \propto \exp\left[-\frac{1}{2\tau^2}\|\boldsymbol{w}\|_2^2\right].$$

Why does the Gaussian prior become an ℓ_2 penalty?

The joint prior assigns lower probability to weight vectors with a larger squared norm:

$$\|\boldsymbol{w}\|_2^2 = w_1^2 + \cdots + w_d^2.$$

Therefore,

$$\|\boldsymbol{w}\|_2^2 \uparrow \quad \Rightarrow \quad p(\boldsymbol{w}) \downarrow .$$

To connect this probability with an optimization objective, take the negative logarithm of the prior density:

$$-\log p(\boldsymbol{w}) = \frac{1}{2\tau^2}\|\boldsymbol{w}\|_2^2 + \frac{d}{2}\log(2\pi\tau^2).$$

The second term does not depend on $\boldsymbol{w}$, so write it as a constant C:

$$-\log p(\boldsymbol{w}) = \frac{1}{2\tau^2}\|\boldsymbol{w}\|_2^2 + C.$$

Maximizing the prior probability is therefore equivalent to minimizing

$$\boxed{\frac{1}{2\tau^2}\|\boldsymbol{w}\|_2^2}.$$

From the Gaussian prior to a MAP objective

We have already seen how the Gaussian prior prefers smaller regression weights. Now combine this prior with the observed regression data. Under Gaussian observation noise,

$$\boldsymbol{y} \mid \boldsymbol{w} \sim \mathcal{N}(\boldsymbol{X}\boldsymbol{w}, \sigma^2 \boldsymbol{I}).$$

MAP selects the weight vector where the posterior density is highest:

$$\hat{\boldsymbol{w}}_{\mathrm{MAP}} = \arg\max_{\boldsymbol{w}} p(\boldsymbol{w} \mid \boldsymbol{y}, \boldsymbol{X}).$$

Using Bayes' rule,

$$p(\boldsymbol{w} \mid \boldsymbol{y}, \boldsymbol{X}) \propto p(\boldsymbol{y} \mid \boldsymbol{X}, \boldsymbol{w}) p(\boldsymbol{w}).$$

Gaussian MAP becomes ridge regression

Taking the negative logarithm turns posterior maximization into cost minimization. The Gaussian likelihood contributes

$$-\log p(\boldsymbol{y} \mid \boldsymbol{X}, \boldsymbol{w}) = \frac{1}{2\sigma^2} \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{w}\|_2^2 + C.$$

The Gaussian prior contributes

$$-\log p(\boldsymbol{w}) = \frac{1}{2\tau^2} \|\boldsymbol{w}\|_2^2 + C.$$

Therefore MAP minimizes

$$\frac{1}{2\sigma^2} \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{w}\|_2^2 + \frac{1}{2\tau^2} \|\boldsymbol{w}\|_2^2.$$

Multiplying by $2\sigma^2$ gives

$$\boxed{\hat{\boldsymbol{w}}_{\text{MAP}} = \arg\min_{\boldsymbol{w}} \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{w}\|_2^2 + \lambda \|\boldsymbol{w}\|_2^2}, \qquad \lambda = \frac{\sigma^2}{\tau^2}.$$

This is the ridge regression objective.

What does the ridge objective mean?

Ridge balances two competing objectives:

$$\underbrace{\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{w}\|_2^2}_{\text{fit the observed data}} + \lambda \underbrace{\|\boldsymbol{w}\|_2^2}_{\text{discourage large weights}} .$$

If λ is small, the solution focuses mainly on fitting the data. If λ is large, keeping the weights small becomes more important. In the Bayesian interpretation,

$$\lambda = \frac{\sigma^2}{\tau^2}.$$

Therefore:

- larger σ^2: noisier observations $\Rightarrow$ rely relatively more on the prior;
- smaller τ^2: stronger belief in small weights $\Rightarrow$ stronger regularization;
- larger τ^2: weaker prior $\Rightarrow$ solution moves closer to ordinary least squares.

Why can ridge make regression more stable?

When features are highly correlated, different weight combinations can produce very similar predictions. As a result, the data may not determine the weights reliably:

$$\text{small change in data} \quad \Rightarrow \quad \text{large change in fitted weights.}$$

Ordinary least squares depends on

$$\boldsymbol{X}^\top \boldsymbol{X},$$

which can become nearly singular when features are redundant. Ridge replaces it with

$$\boldsymbol{X}^\top \boldsymbol{X} + \lambda \boldsymbol{I}.$$

This moves small eigenvalues away from zero, making the solution better determined and less sensitive to noise.

Prior choice determines the regularization penalty

MAP minimizes the negative log-posterior, so the prior enters the objective through

$$-\log p(\boldsymbol{w}).$$

Different prior shapes therefore produce different penalties:

Prior on w	MAP penalty	Effect on weights
Gaussian	w^2 (ℓ_2)	smoothly shrinks weights toward zero
Laplace	$\lvert w \rvert$ (ℓ_1)	can drive some weights exactly to zero
Uniform on a set	hard constraint	forbids values outside the allowed set

For example, the Gaussian prior decreases smoothly as $|w|$ grows, so large weights receive an increasingly large quadratic cost.

What should we learn from MAP and ridge?

The Gaussian case gives a complete connection:

$$\boxed{\text{Gaussian prior on } \boldsymbol{w} \to -\log p(\boldsymbol{w}) \to \ell_2 \text{ penalty} \to \text{ridge regression}}.$$

This connection has two interpretations:

- **Bayesian:** large weights are less plausible under the chosen prior;
- **regularization:** shrinking poorly supported weights can make estimation more stable.

However, MAP keeps only the most probable parameter vector:

$$\hat{\boldsymbol{w}}_{\mathrm{MAP}} = \arg\max_{\boldsymbol{w}} p(\boldsymbol{w} \mid \mathcal{D}).$$

Full Bayesian inference instead keeps

$$p(\boldsymbol{w} \mid \mathcal{D}),$$

which describes uncertainty over many possible parameter values.

Interpretation and limitations

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MLE, MAP, and full Bayesian inference

Method	Output	Uses prior?	Retains parameter uncertainty?
MLE	point estimate	no	no
MAP	point estimate	yes	no
Posterior mean	point summary	yes	no
Full posterior	distribution	yes	yes

The posterior mean is computed from the full posterior, but once it is reduced to one number, information about posterior spread and shape is lost. Posterior prediction can retain this uncertainty by averaging predictions over the full posterior.

When priors help and when they hurt

Priors help when they:

- encode genuine previous evidence or structural knowledge;
- stabilize weakly identified or high-dimensional models;
- prevent impossible or unreasonable parameter values.

Priors hurt when they are strong, wrong, and not stress-tested.

Good practice

Report the prior, inspect sensitivity to plausible alternatives, and check prior predictive behavior.

Summary

1. Bayes' rule updates a parameter distribution using observed data.
2. A conjugate prior keeps the posterior calculation in the same distribution family.
3. Posterior prediction averages over parameter uncertainty.
4. The maximum a posteriori estimate combines data fit with prior information.
5. A Gaussian prior produces a squared penalty on large parameters.

Next week: group unlabeled observations by alternating between estimated cluster membership and model updates.

Bayesian learning balances evidence and prior structure

Questions and discussion